

HIV, AIDS

(See also *Harrison's Principles of Internal Medicine*, 17th Edition, Chapter 182)

Definition

- HIV disease
 - An infectious disease caused by HIV, a human retrovirus
 - HIV disease should be viewed as a spectrum ranging from primary infection, with or without the acute syndrome, to an asymptomatic stage, to advanced disease characterized by profound immunodeficiency and susceptibility to opportunistic infections.
- AIDS
 - Late stage of infection with HIV
 - Current case definition
 - Any HIV-infected person with a CD4+ T-cell count $<200/\mu\text{L}$
 - Development of an AIDS-defining clinical condition (see Classification)

Epidemiology

- Prevalence worldwide
 - A global pandemic, with cases reported from virtually every country
 - ~38 million adults were living with HIV/AIDS as of the end of 2005.
 - Two-thirds of these adults are in sub-Saharan Africa.
 - ~50% are women.
 - ~2.3 million children <15 years are living with HIV/AIDS.
 - In 2005, there were ~5 million new cases worldwide.
 - Through 2005, the cumulative number of AIDS-related deaths worldwide exceeds 25 million.
 - HIV/AIDS is the second leading infectious cause of death worldwide.
- Prevalence in the U.S.
 - ~1 million people were living with HIV/AIDS as of the end of 2004.
- Incidence in the U.S.
 - The number of new infections per year is estimated to be ~40,000.
 - This number has remained stable for more than a decade.
 - Among persons newly diagnosed with HIV/AIDS from 2001–2004 in U.S. states with confidential name-based reporting:
 - ~71% are male.
 - The Centers for Disease Control and Prevention estimate that ~60% were infected through male-to-male sexual contact.
 - 16% through injection drug use
 - 5% through male-to-male sexual contact and injection drug use
 - 17% through heterosexual transmission
 - ~29% are female.
 - ~76% were infected through heterosexual transmission.
 - 21% through injection drug use

- o New diagnoses of HIV/AIDS
 - Decreased approximately 2% among all men and 15% among all women from 2001 to 2004
 - Increased among men who have sex with men, but decreased among men and women in other risk categories
- Distribution (U.S.)
 - o Most cases of transmission by injection drug use and heterosexual transmission are reported from the Northeast and Southeast, particularly among minority groups.
 - o HIV infection and AIDS have disproportionately affected minority groups in the U.S.
 - o The estimated rates of AIDS diagnoses per 100,000 persons among adults and adolescents in 2004 were:
 - 72.1 for African Americans
 - 25.0 for Hispanics
 - 7.1 for whites
 - 9.9 for American Indians/Alaska Natives
 - 4.4 for Asian/Pacific Islanders

Risk Factors

- **Sexual transmission**
 - o Homosexual and heterosexual contact with an infected person
 - o In U.S.
 - 44% of new HIV/AIDS diagnoses in 2001–2004 were attributed to male-to-male sexual contact.
 - 34% of new HIV/AIDS diagnoses in 2001–2004 were attributed to heterosexual contact.
 - o Worldwide
 - Heterosexual transmission is the most common mode of infection.
 - o Male-to-female transmission is 8 times more efficient than female to male.
 - o Receptive anal intercourse is a much more efficient mode of transmission than oral sex.
 - o The presence of other sexually transmitted diseases significantly increases the risk of transmission, especially those with genital ulceration.
 - o Lack of circumcision carries an increased risk of HIV infection.
 - o The association of alcohol consumption and illicit drug use with unsafe sexual behavior leads to an increased risk of sexual transmission of HIV.
- Transmission by blood and blood products
 - o Transmission by HIV-tainted blood transfusions, blood products, or transplanted tissue
 - o Intravenous drug users
 - Exposed to HIV while sharing injection paraphernalia, such as needles, syringes, the water in which the drugs are mixed, or the cotton through which drugs are filtered
 - Does not require intravenous puncture
 - Subcutaneous (skin popping) or intramuscular (muscling) injections can transmit HIV.
- Occupational transmission of HIV (health care workers and laboratory personnel)
 - o Risk of HIV transmission after skin puncture from a needle or a sharp object that was contaminated with blood from a person with documented HIV infection is ~0.3%, and after a mucous membrane exposure it is 0.09%.
 - o Transmission after nonintact skin exposure has been documented.

- o The risk is estimated to be less than the risk for mucous membrane exposure.
- o Risk of transmission from an infected health care worker to patients is extremely low; in fact, too low to be measured accurately.
- Maternal-fetal/infant transmission
 - o Can be transmitted intrapartum, perinatally (most commonly), or via breast milk
 - o In the absence of prophylactic antiretroviral therapy to the mother during pregnancy, labor, and delivery, and to the fetus following birth, the probability of transmission of HIV from mother to infant/fetus ranges from:
 - 15–25% in industrialized countries
 - 25–35% in developing countries
- Transmission by other body fluids
 - o Although the virus can be identified from virtually any body fluid, there is no evidence that HIV can be transmitted as a result of exposure to saliva, tears, sweat, or urine.
 - o Transmission of HIV by a human bite can occur but is rare.

Etiology

Human retroviruses HIV-1 and HIV-2

- Family of human retroviruses (Retroviridae)
- Subfamily of lentiviruses
- RNA viruses whose hallmark is the reverse transcription of its genomic RNA to DNA by the enzyme reverse transcriptase
- HIV-1 is the most common cause of AIDS worldwide.
- HIV-2 has been identified predominantly in western Africa.
 - o Small numbers of cases have also been reported in Europe, South America, Canada, and the U.S.
 - o Has ~40% sequence homology with HIV-1
 - o More closely related to simian immunodeficiency viruses

Transmission

- See Risk Factors.
- Sexual contact: heterosexual and homosexual
- Contact with blood, blood products, or other bodily fluids (as in drug abusers who share contaminated intravenous needles)
- Intrapartum or perinatally from mother to infant or via breast milk
- Contact with HIV-infected specimens (a definite but small occupational risk of infection for health care workers and laboratory personnel)

Pathophysiology and immunopathogenesis

- Hallmark of HIV disease is a profound immunodeficiency.
- Results from a progressive deficiency of the subset of T lymphocytes (CD4+ T cells), referred to as *helper* or *inducer T cells*.
 - o The CD4 molecule serves as the primary cellular receptor for HIV.
 - o A co-receptor must be present with CD4 for efficient entry of HIV-1 into target cells.
 - o The 2 major co-receptors for HIV-1 are CCR5 and CXCR4.
- Although the CD4+ T lymphocyte and CD4+ monocyte lineage are the principal cellular targets of HIV, virtually any cell that expresses CD4 along with one of the co-receptors can potentially be infected by HIV.

Primary infection

- After initial transmission, the virus infects CD4+ cells, probably T lymphocytes, monocytes, or bone marrow-derived dendritic cells.
- Ultimately, lymph node architecture is completely disrupted and the efficiency of trapping virions declines, leading to equilibration of the viral burden between peripheral blood cells and lymph node cells.
- Most patients undergo a viremic stage during primary infection.
 - In some, this is associated with the "acute retroviral syndrome," a mononucleosis-like illness.
 - Virus is disseminated to lymphoid and other organs throughout the body.
 - Infection is ultimately contained partially by the development of an HIV-specific immune response and the trapping of virions in lymphoid tissue.

Establishment of chronic and persistent infection

- Despite the robust immune response following primary infection, virus is not cleared from the body.
 - Instead, a chronic infection develops that persists for a median of 10 years before the patient becomes clinically ill.
- During this period of clinical latency
 - The number of CD4+ T cells gradually decreases.
 - Few, if any, clinical findings are evident.
 - Active viral replication can almost always be detected by measurable plasma viremia and the demonstration of virus replication in lymphoid tissue.
- Level of steady-state viremia (referred to as the *viral set point*) at approximately 1 year after infection
 - Has important prognostic implications for the progression of HIV disease
 - Persons with a low viral set point at 6 months to 1 year after infection progress to AIDS more slowly than those whose set point is very high at this time.

Advanced HIV disease

- When CD4+ T-cell counts will fall below a critical level ($\sim 200/\mu\text{L}$), patients become highly susceptible to opportunistic disease.
- Control of plasma viremia by effective antiretroviral therapy
 - Can increase survival even in persons with extremely low CD4+ T-cell counts
 - Even though CD4+ T-cell counts may not increase significantly as a result of therapy

Symptoms & Signs

Acute HIV syndrome

- 50–70% of persons with HIV infection experience an acute clinical syndrome approximately 3–6 weeks after primary infection.
- Usually persists for 1 to several weeks
- General
 - Fever
 - Pharyngitis
 - Lymphadenopathy (70% of cases)
 - Headache/retro-orbital pain
 - Arthralgias/myalgias

- o Lethargy/malaise
- o Anorexia
- o Weight loss
- o Nausea
- o Vomiting
- o Diarrhea
- Neurologic
 - o Meningitis
 - o Encephalitis
 - o Peripheral neuropathy
 - o Myelopathy
- Dermatologic
 - o Erythematous maculopapular rash
 - o Mucocutaneous ulceration

Asymptomatic infection

- The length of time between infection and development of disease varies, but the median is ~10 years.
- Active viral replication continues during this asymptomatic period, and CD4+ T-cell counts decrease.
- Rate of disease progression is directly correlated with plasma HIV RNA levels.

Symptomatic disease

- Symptoms can develop at any time during the course of HIV infection.
 - o A spectrum of illness generally changes as the CD4+ T-cell count decreases.
 - o More severe and life-threatening complications of HIV infection occur in patients with a CD4+ T-cell count <200/ μ L.
- Major clinical syndromes seen in the symptomatic stage of HIV infection are summarized below.
- Persistent generalized lymphadenopathy
 - o Palpable adenopathy at ≥ 2 extrainguinal sites that persists for >3 months without explanation other than HIV infection
- Constitutional symptoms
 - o Fever persisting for >1 month
 - o Involuntary weight loss of >10% of baseline
 - o Diarrhea for >1 month in absence of explainable cause
- Neurologic disease
 - o HIV encephalopathy (AIDS dementia complex): most common
 - o Opportunistic infections
 - o Primary central nervous system (CNS) lymphoma
 - o CNS Kaposi's sarcoma
 - o Aseptic meningitis
 - o Myelopathy
 - o Peripheral neuropathy
 - o Myopathy
- Secondary infectious diseases
 - o *Pneumocystis carinii* pneumonia
 - o most common opportunistic infection, occurring in ~80% of individuals during the course of their illness.
 - o Cytomegalovirus (CMV) (chorioretinitis, colitis, pneumonitis, adrenalitis)

- o *Candida albicans* (oral thrush, esophagitis)
 - o *Mycobacterium avium* intracellulare (localized or disseminated infection)
 - o *Mycobacterium tuberculosis*
 - o *Cryptococcus neoformans* (meningitis, disseminated disease)
 - o *Toxoplasma gondii* (encephalitis, intracerebral mass lesion)
 - o Herpes simplex virus (severe mucocutaneous lesions, esophagitis)
 - o Diarrhea due to *Cryptosporidium* species or *Isospora belli*
 - o Bacterial pathogens (especially in pediatric cases)
- Secondary neoplasms
 - o Kaposi's sarcoma (cutaneous and visceral, more fulminant course than in non-HIV-infected patients)
 - o Lymphoid neoplasms (especially B cell lymphomas of brain, marrow, GI tract)
- Organ-specific disease
 - o A variety of organ-specific manifestations and complications can be seen, either as primary manifestations of the HIV infection or as complications of treatment (see Complications).

Differential Diagnosis

- Acute HIV syndrome
 - o Epstein-Barr virus (mononucleosis)
 - o Acute Hepatitis A or B
 - o Roseola
 - o Secondary syphilis
 - o Cytomegalovirus
 - o Toxoplasmosis
 - o Other acute viral syndromes
- Symptomatic disease
 - o Depends on which of many HIV complications may be causing symptoms

Diagnostic Approach

- Diagnosis of HIV infection
 - o Diagnosis depends on demonstration of antibodies to HIV and/or direct detection of HIV or 1 of its components.
 - o Antibodies to HIV appear 2–12 weeks after infection.
- Initial evaluation of patient with HIV infection
 - o History and physical examination
 - o Routine chemistry and hematology
 - o CD4+ T-lymphocyte count
 - o 2 plasma HIV RNA levels
 - o Rapid plasma reagin test
 - o Anti-*Toxoplasma* antibody titer
 - o Purified protein derivative skin test
 - o Mini-Mental Status Examination
 - o Serologies for hepatitis A, B, and C
 - o Immunization with pneumococcal polysaccharide; influenza as indicated
 - o Immunization with hepatitis A and hepatitis B if seronegative
 - o Counseling regarding natural history and transmission
 - o Helping contact others who might be infected

- Diagnosis of AIDS
 - CD4+ T-cell count <200/ μ L **or**
 - AIDS-defining clinical condition(see Classification)

Laboratory Tests

Diagnosis of HIV infection

- See Figure 1 for the algorithm of serologic testing.

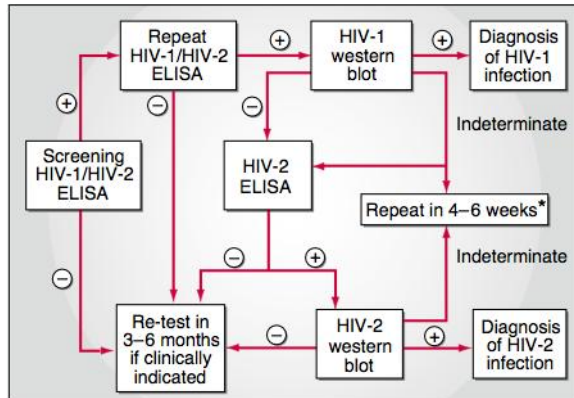


Figure 1: Algorithm for the use of serologic tests in the diagnosis of HIV-1 or HIV-2 infection.

*Stable indeterminate western blot 4 to 6 weeks later makes HIV infection unlikely. However, it should be repeated twice at 3-month intervals to rule out HIV infection. Alternatively, one may test for HIV-1 p24 antigen or HIV RNA.

- Enzyme immunoassay (EIA) (enzyme-linked immunosorbent assay)
 - Standard screening test for HIV infection
 - The test is highly sensitive (>99.5%).
 - Most commercial EIA kits can detect antibodies to both HIV-1 and -2.
 - Test not optimally specific, therefore confirmatory test needed
 - False-positive results can occur with:
 - Antibodies to class II antigens
 - Autoantibodies
 - Hepatic disease
 - Recent influenza vaccination
 - Acute viral infections
- Western blot
 - Most commonly used confirmatory test
 - Detects antibodies to HIV antigens of specific molecular weights
 - Antibodies to HIV begin to appear within 2 weeks of infection.
 - Period of time between initial infection and development of detectable antibodies is rarely >3 months.
- Tests for direct detection of HIV
 - These tests are useful in:
 - Patients with a positive or indeterminate EIA result and an indeterminate Western blot result **or**
 - Patients in whom serologic testing may be unreliable (such as those with hypogammaglobulinemia)

- o Immune complex dissociated p24 antigen capture assay
 - Plasma p24 antigen levels increase during the first few weeks following infection, before the appearance of anti-HIV antibodies.
- o HIV RNA by polymerase chain reaction
- o HIV RNA by branched DNA
- o HIV RNA by nucleic acid sequence-based assay

Other tests at diagnosis

- Complete blood count
- Chemistry profile
- Aminotransferase measurement
- Blood urea nitrogen and creatinine measurement
- Urinalysis
- Pap smear
- Fasting blood glucose and serum lipid measurement

Monitoring HIV infection and response to therapy

- CD4+ T-cell count
 - o Generally accepted indicator of immunologic competence
 - o Close relationship between the CD4+ count and clinical manifestations of AIDS
 - $<200/\mu\text{L}$: high risk of infection with *Pneumocystis carinii*
 - $<50/\mu\text{L}$: high risk for CMV disease and infection with *Mycobacterium avium intracellulare*
 - o Should be measured:
 - At the time of diagnosis
 - Every 3–6 months thereafter
 - Measurements may be done more frequently in patients with decreasing counts.
 - o Antiretroviral therapy
 - A CD4+ T-cell count $<250/\mu\text{L}$ is an indication to start therapy according to most practice guidelines.
- HIV RNA level
 - o Predicts what will happen to the CD4+ T cell count in the near future and may itself be correlated with immune dysfunction
 - o Should be measured:
 - At the time of HIV diagnosis
 - Every 3–4 months thereafter in the untreated patient
 - o Useful in making therapeutic decisions about antiretroviral therapy
 - After initiation of therapy or any change in therapy, HIV RNA levels should be monitored approximately every 4 weeks until the effectiveness of the therapeutic regimen is determined by development of a new steady-state level of HIV RNA.
 - During therapy, levels of HIV RNA should be monitored every 3–4 months to evaluate the continuing effectiveness of therapy.
- HIV resistance testing
 - o Sensitivity of patient's HIV virus(es) to different antiretroviral agents can be tested by genotypic or phenotypic assays.
 - o In the hands of experts, the use of resistance testing enhances the short-term ability to achieve ~ 0.5 -log decreases in viral load compared to changing drugs merely on the basis of drug history.

- o The clinical value of HIV resistance testing in selecting an initial treatment regimen is still under investigation.

Imaging

- Directed by specific HIV complications

Diagnostic Procedures

- See Diagnostic Approach and Laboratory Tests.

Classification

Clinical categories of HIV infection

Category A

- ≥ 1 of the conditions listed below in an adolescent or adult (>13 years) with documented HIV infection; conditions seen in B and C must not have occurred
 - o Asymptomatic HIV infection
 - o Persistent generalized lymphadenopathy
 - o Acute (primary) HIV infection with accompanying illness or history of acute HIV infection

Category B (symptomatic)

- Symptomatic conditions in an HIV-infected adolescent or adult that are not included among conditions listed in clinical category C and that meet ≥ 1 of the following criteria:
 - o Conditions are attributed to HIV infection or are indicative of a defect in cell-mediated immunity **or**
 - o Conditions are considered by physicians to have a clinical course or to require management complicated by HIV infection
- Examples include, but are not limited to:
 - o Bacillary angiomatosis
 - o Candidiasis, oropharyngeal (thrush)
 - o Candidiasis, vulvovaginal: persistent, frequent, or poorly responsive to therapy
 - o Cervical dysplasia (moderate or severe)/cervical carcinoma in situ
 - o Constitutional symptoms, such as fever (38.5°C) or diarrhea lasting >1 month
 - o Hairy leukoplakia, oral
 - o Herpes zoster (shingles), involving ≥ 2 distinct episodes or >1 dermatome
 - o Idiopathic thrombocytopenic purpura
 - o Listeriosis
 - o Pelvic inflammatory disease, particularly if complicated by tubo-ovarian abscess
 - o Peripheral neuropathy

Category C (AIDS)

- Conditions listed in the AIDS surveillance case definition
 - o Candidiasis of bronchi, trachea, or lungs
 - o Candidiasis, esophageal
 - o Cervical cancer, invasive

- o Coccidioidomycosis, disseminated or extrapulmonary
- o Cryptococcosis, extrapulmonary
- o Cryptosporidiosis, chronic intestinal (>1 month's duration)
- o Cytomegalovirus disease (other than liver, spleen, or nodes)
- o Cytomegalovirus retinitis (loss of vision)
- o Encephalopathy (HIV-related)
- o Herpes simplex: chronic ulcer(s) (>1 month's duration) or bronchitis, pneumonia, or esophagitis
- o Histoplasmosis, disseminated or extrapulmonary
- o Isoporiasis, chronic intestinal (>1 month's duration)
- o Kaposi's sarcoma
- o Lymphoma, Burkitt's (or equivalent term)
- o Lymphoma, primary, of brain
- o *M. avium* complex, or *Mycobacterium kansasii*, disseminated or extrapulmonary
- o *Mycobacterium tuberculosis*, any site (pulmonary or extrapulmonary)
- o *Mycobacterium*, other species or unidentified species, disseminated or extrapulmonary
- o *P. carinii* pneumonia
- o Pneumonia, recurrent
- o Progressive multifocal leukoencephalopathy
- o *Salmonella* septicemia, recurrent
- o Toxoplasmosis of brain
- o Wasting syndrome due to HIV

Treatment Approach

- Principles of therapy of HIV infection
 - o Ongoing HIV replication leads to immune system damage and progression to AIDS.
 - o Plasma HIV RNA levels indicate the magnitude of HIV replication and the rate of CD4+ T-cell destruction.
 - o CD4+ T-cell counts indicate the current level of competence of the immune system.
 - o Rates of disease progression differ among patients, and treatment decisions should be individualized based on plasma HIV RNA levels and CD4+ T-cell counts.
 - o Maximal suppression of viral replication is a goal of therapy; the greater the suppression, the less likely the appearance of drug-resistance quasiespecies.
 - o The most effective therapeutic strategies involve simultaneous initiation of combinations of anti-HIV drugs active against the patient's virus.
 - o The antiretroviral drugs used in combination regimens should be used according to optimum schedules and dosages.
 - o The number of available drugs is limited; any decisions on antiretroviral therapy have a long-term impact on future options for the patient.
 - o Women should receive optimal antiretroviral therapy regardless of pregnancy status.
 - o Compliance is an important part of ensuring maximal effect from a given regimen.
 - The simpler the regimen, the easier it is for the patient to be compliant.
- Indications for initiation of antiretroviral therapy
 - o Acute infection syndrome (no clear consensus)
 - o Chronic infection
 - Symptomatic disease
 - Asymptomatic diseases
 - CD4+ T-cell count <250/ μ L or decreasing
 - HIV RNA >50,000 copies/mL or increasing
 - o Postexposure prophylaxis
- Indications for changing antiretroviral regimens

- o <1-log decrease in plasma HIV RNA by 4 weeks after initiation of therapy
- o A reproducible significant increase (defined as 3-fold or greater) from the nadir of plasma HIV RNA level not attributable to intercurrent infection, vaccination, or test method
- o Persistently decreasing CD4+ T-cell count
- o Clinical deterioration
- o Side effects
- Treatment prophylaxis considerations based on CD4+ count
 - o <200/ μ L
 - *P. carinii* prophylaxis
 - o <50/ μ L
 - Primary prophylaxis for *M. avium* infection

Specific Treatments

Antiretroviral therapy

- Combination highly active antiretroviral therapy is the cornerstone of management of patients with HIV infection
 - o Marked decrease in incidence of most AIDS-defining conditions since initiation of widespread use of highly active antiretroviral therapy in the U.S. in 1995–1996
- Primary goals of treatment
 - o Reduce HIV-related morbidity and mortality.
 - o Improve quality of life.
 - o Restore and preserve immunologic function.
 - o Maximally and durably suppress viral load.
- Indications for treatment
 - o All patients with history of an AIDS-defining illness or severe symptoms of HIV infection regardless of CD4+ T-cell count.
 - o Asymptomatic patients with <250 CD4+ T cells/ μ L
 - o For asymptomatic patients with a CD4+ T-cell count >250 cells/ μ L and plasma HIV RNA level >100,000 copies/mL, most experienced clinicians defer therapy, but some clinicians may consider initiating treatment.
 - o Therapy should be deferred for patients with CD4+ T-cell counts >350 cells/ μ L and plasma HIV RNA level <100,000 copies/mL.
 - o Treatment for acute and recent HIV infection is considered optional.
 - Long-term virologic, immunologic, or clinical benefit is unknown.
- Antiretroviral drugs
 - o Nucleoside/nucleotide reverse transcriptase inhibitors
 - Zidovudine (Retrovir)
 - Didanosine (Videx, Videx EC)
 - Zalcitabine (HIVID)
 - Stavudine (Zerit)
 - Lamivudine (Epivir)
 - Emtricitabine (FTC, Emtriva)
 - Abacavir (Ziagen)
 - Tenofovir (Viread)
 - o Non-nucleoside reverse transcriptase inhibitors
 - Efavirenz (Sustiva)
 - Nevirapine (Viramune)
 - Delavirdine (Rescriptor)

- o Protease inhibitors
 - Lopinavir/ritonavir (Kaletra)
 - Atazanavir (Reyataz)
 - Fosamprenavir (Lexiva)
 - Saquinavir mesylate (Invirase—hard gel capsule; Fortovase—soft gel capsule)
 - Ritonavir (Norvir)
 - Indinavir sulfate (Crixivan)
 - Nelfinavir mesylate (Viracept)
 - Amprenavir (Agenerase)
- o Fusion inhibitor
 - Enfuvirtide (Fuzeon)
- Preferred regimens for treatment-naïve patients (optimal regimen remains in flux)
 - o Non-nucleoside reverse transcriptase inhibitor based
 - Efavirenz **plus** (lamivudine or emtricitabine) **plus** (zidovudine or tenofovir DF)
 - Except during first trimester of pregnancy or in women with high pregnancy potential
 - o Protease inhibitor based
 - Atazanavir **or** lopinavir/ritonavir (coformulation) **plus** (lamivudine or emtricitabine) **plus** (zidovudine or tenofovir DF)
 - o For more details and alternative regimens, see the U.S Department of Health and Human Services Guidelines for the Use of Antiretroviral Agents in HIV-1-Infected Adults and Adolescents.
- Adverse effects and drug–drug interactions are common and must be taken into consideration when using these medications.
 - o One of the main problems has been a syndrome of hyperlipidemia and fat redistribution referred to as *lipodystrophy syndrome*.

Virologic failure during treatment

- Virologic failure is currently defined as:
 - o HIV RNA level >400 copies/mL after 24 weeks
 - o >50 copies/mL after 48 weeks
 - o Repeated HIV RNA level >400 copies/mL after prior suppression of viremia to <400 copies/mL
- Evaluation of antiretroviral treatment failure should include:
 - o Assessing the severity of HIV disease
 - o Antiretroviral treatment history, including the duration, drugs used, antiretroviral potency, adherence history, and drug intolerance/toxicity
 - o Results of prior drug resistance testing
- Drug resistance testing
 - o Should be done while the patient is taking the failing antiretroviral regimen (or as soon as possible within 4 weeks of treatment discontinuation)
- Managing virologic failure
 - o Use the treatment history and past and current resistance test results to identify active agents (preferably at least two fully active agents) to design a new regimen.
 - A fully active agent is one likely to demonstrate antiretroviral activity on the basis of both the treatment history and susceptibility on drug-resistance testing.
 - o If at least 2 fully active agents cannot be identified:
 - Consider pharmacokinetic enhancement of protease inhibitors (with the exception of nelfinavir) with ritonavir **and/or**
 - Reuse other prior antiretroviral agents to provide partial antiretroviral activity.

- o Significant antiretroviral activity can be obtained by adding to an optimized background antiretroviral regimen.
 - A drug with activity against drug-resistant virus (e.g., a potent ritonavir-boosted PI)
 - A drug with a new mechanism of action (e.g., HIV entry inhibitor)
- o One active drug should not be added to a failing regimen because drug resistance is likely to develop quickly.
 - However, in patients with advanced HIV disease (e.g. CD4+ T-cell count <100/ μ L) and high risk of clinical progression, adding 1 active agent (with an optimized background regimen) may provide clinical benefits and should be considered.

Prevention of opportunistic infections

Primary and secondary prophylaxis

- *P. carinii*
 - o Indications
 - CD4 count <200/ μ L **or**
 - Oropharyngeal candidiasis **or**
 - Unexplained fever >2 weeks **or**
 - Prior bout of *P. carinii* pneumonia
 - May stop prophylaxis if CD4+ T-cell count >200/ μ L for 6 months
 - o First choice
 - Trimethoprim-sulfamethoxazole, 1 double-strength tablet qd
- *M. tuberculosis*
 - o Indications
 - Skin test >5 mm **or**
 - Prior positive test without treatment **or**
 - Close contact with case of active tuberculosis
 - o First choices
 - Isoniazid, 300 mg PO qd **plus** pyridoxine, 50 mg PO qd for 9 months
 - Isoniazid, 900 mg PO **plus** pyridoxine, 100 mg PO twice weekly for 9 months
 - o If drug resistance is suspected, check with local public health authorities.
- *M. avium* complex
 - o Indications
 - CD4 count <50/ μ L
 - Prior documented disseminated disease
 - May stop prophylaxis if CD4+ T-cell count >100/ μ L for 6 months
 - o First choice
 - Azithromycin: 1,200 mg PO weekly

Immunizations recommended

- Hepatitis B virus
 - o Indications
 - All susceptible (anti-hepatitis B core antigen and anti-hepatitis B surface antigen negative) patients

- Hepatitis A virus
 - Indications
 - All susceptible (anti-hepatitis A virus negative) patients at increased risk for hepatitis A
- Influenza virus
 - Indications
 - All patients annually
- *Streptococcus pneumoniae*
 - Indications
 - All patients

Recommendations for prevention of severe and frequent recurrences

- Herpes simplex
 - Indications
 - Frequent/severe recurrences
 - First choices
 - Acyclovir: 200 mg PO tid
 - Acyclovir: 400 mg PO bid
 - Famciclovir: 250 mg PO bid
- *Candida*
 - Indications
 - Frequent/severe recurrences
 - First choice
 - Fluconazole: 100–200 mg PO qd

Monitoring

- CD4+ T-cell counts
 - Measure every 3–6 months after diagnosis.
 - Measurements may be done more frequently in patients with decreasing counts.
 - <200/ μ L: high risk for infection with *P. carinii*
 - <50/ μ L: at high risk of infection with CMV, *M. avium* complex
- Plasma HIV RNA levels
 - Useful in making decisions about antiretroviral therapy
 - In untreated patients
 - Measure every 3–6 months after diagnosis.
 - After initiation or any change in therapy
 - Monitor about every 4 weeks until effectiveness is determined by development of new steady-state level of HIV RNA.
 - During chronic therapy
 - Monitor every 3–4 months to evaluate the continuing effectiveness of therapy.

Complications

General

- For details, see U.S. Health and Human Services Treating Opportunistic Infections Among HIV-Infected Adults and Adolescents.

Diseases of the respiratory system

- Acute bronchitis
- Sinusitis
- Pneumonia
 - Bacterial
 - Tuberculosis
 - Atypical mycobacterial infections
 - *Rhodococcus equi* infection
- Fungal infections of the lung
 - Coccidioidomycosis
 - Aspergillus infection
 - Histoplasmosis
- Idiopathic interstitial pneumonia

Diseases of the cardiovascular system

- HIV-associated cardiomyopathy
- Drug-induced cardiomyopathy
- Cardiomyopathy due to:
 - Kaposi's sarcoma
 - Cryptococcosis
 - Chagas' disease
 - Toxoplasmosis
- Pericardial effusions
- Pericarditis
- Nonbacterial thrombotic endocarditis
- Ischemic heart disease/myocardial infarction

Diseases of the oropharynx and GI system

- Most frequently due to secondary infections
- Thrush
- Hairy leukoplakia
- Aphthous ulcers
- Esophagitis
 - Candida
 - CMV
 - Herpes simplex virus
- Infections of the small and large intestines
 - Bacterial (Salmonella, Shigella, Campylobacter)
 - Fungal (histoplasmosis, coccidioidomycosis, penicilliosis)
 - Protozoal (cryptosporidia, microsporidia, *Isospora belli*)
- Peritonitis
- CMV colitis
- AIDS enteropathy
- Rectal lesions
 - Herpes simplex virus
 - Condylomata acuminata
 - Kaposi's sarcoma
 - Intraepithelial neoplasia

Hepatobiliary disease

- Hepatitis B
- Hepatitis C
- Granulomatous hepatitis
- Hepatic masses
- Biliary tract disease
 - Papillary stenosis
 - Sclerosing cholangitis
- Liver injury due to drug toxicity
- Pancreatic injury due to drug toxicity

Diseases of the kidney and genitourinary tract

- HIV-associated nephropathy
- Drug-induced renal damage
- Genitourinary tract infections
- Infections with *Treponema pallidum*
- Vulvovaginal candidiasis
- Vaginitis

Diseases of the endocrine system and metabolic disorders

- Lipodystrophy
 - Elevated triglyceride level
 - Elevated total cholesterol level
 - Elevated lipoprotein B level
 - Hyperinsulinemia
 - Hyperglycemia
- Avascular necrosis of the hips and shoulders
- Lactic acidosis
- Hyponatremia
 - Syndrome of inappropriate antidiuretic hormone secretion
 - Adrenal insufficiency
- Hypothyroidism
- Hypogonadism

Rheumatologic diseases

- Immune reactivation syndromes
 - After initiation of antiretroviral therapy, an exaggerated immune response to opportunistic infections
- Drug allergies
- Diffuse infiltrative lymphocytosis syndrome
- Reactive arthritis
- Psoriatic arthritis
- AIDS-associated arthropathy
- Painful articular syndrome
- Fibromyalgia
- Leukocytoclastic vasculitis
- CNS angiitis

- Polymyositis
- Septic arthritis

Diseases of the hepatopoietic system

- Lymphadenopathy
- Anemia
- Leukopenia
- Neutropenia
- Thrombocytopenia
- Persistent generalized lymphadenopathy

Dermatologic diseases

- Seborrheic dermatitis
- Eosinophilic pustular folliculitis
- Norwegian scabies
- Psoriasis
- Ichthyosis
- Reactivation herpes zoster (shingles)
- Herpes simplex virus
- Herpetic whitlow
- Diffuse skin eruptions due to *Molluscum contagiosum*
- *Condyloma acuminatum* lesions
- Erythroderma
- Stevens–Johnson syndrome

Neurologic diseases

- Opportunistic infections
 - Toxoplasmosis
 - Cryptococcosis
 - Progressive multifocal leukoencephalopathy
 - Cytomegalovirus
 - *Mycobacterium tuberculosis*
- Co-infections
 - Syphilis
 - Human T-lymphotropic virus 1 infection
- Neoplasms
 - Primary CNS lymphoma
 - Kaposi's sarcoma
- Result of HIV-1 infection
 - Aseptic meningitis
 - HIV encephalopathy (AIDS dementia complex)
- Myelopathy
 - Vacuolar myelopathy
 - Pure sensory ataxia
 - Paresthesia/dysesthesia
- Peripheral neuropathy
 - Acute inflammatory demyelinating polyneuropathy (Guillain–Barré syndrome)
 - Mononeuritis multiplex
 - Distal symmetric polyneuropathy

- Myopathy
- Seizures
- Focal neurologic deficits
- Progressive multifocal leukoencephalopathy
 - JC virus

Ophthalmologic disease

- Cotton-wool spots
- CMV retinitis
- Progressive outer retinal necrosis syndrome
- *P. carinii*
- Toxoplasmosis

Additional disseminated infections and wasting syndrome

- Bacillary angiomatosis
- Histoplasmosis
- *Penicillium marneffei* infection
- Visceral leishmaniasis
- Generalized wasting
- CNS Chagas' disease

Neoplastic diseases

- Kaposi's sarcoma
- Non-Hodgkin lymphoma
- Primary CNS lymphoma
- Cervical, brain, testicular, oral, lung, and anal cancers

Prognosis

- The median time from primary HIV infection to the development of AIDS in untreated persons in the developed world is ~10 years.
 - This period has been markedly extended by the wide availability of combinations of antiretroviral drugs.
- Mortality
 - In 2003, HIV infection was the sixth leading cause of death among Americans 25–44 years of age in the U.S.
 - The annual number of AIDS-related deaths in the U.S. decreased ~70% from 1995 to 2003.
 - Currently, ~5 deaths annually per 100,000 persons
- Approximately 60% of deaths among patients with AIDS are a direct result of an infection other than HIV, primarily:
 - Viral hepatitis
 - Other non-AIDS-defining bacterial infections

Prevention

- Cornerstones of prevention
 - Education

- o Counseling
- o Behavior modification
 - Abstinence or practicing safer sex (condoms together with the spermicide nonoxynol-9)
 - Stopping the use of injectable drugs or using sterilized needles and no sharing of drug paraphernalia
- Screening of blood products
 - o Risk of transmission by transfusion has reduced to 1 in 725,000 to 1 in 835,000.
 - o Autologous blood donation is a better choice, if possible.
 - o Screening of all blood donors for HIV infection by assays for both HIV antibody and p24 antigen and self-deferral of persons at risk for HIV infection.
 - o Clotting factor concentrates are heat treated, essentially eliminating the risk to hemophiliacs who require these products.
- Screening of donors for:
 - o Artificial insemination
 - o Tissues used in organ transplantation
- Use of universal precautions by all health care workers when caring for all patients
- Current recommendations to reduce perinatal transmission of HIV include:
 - o Universal voluntary HIV testing and counseling of pregnant women
 - o Antiretroviral prophylaxis with ≥ 1 drugs in cases in which the mother does not require therapy for her HIV infection
 - o Combination therapy for women who do require therapy
 - o Obstetric management to minimize exposure of the infant to maternal blood and genital secretions
 - o Avoidance of breast-feeding
- Vaccines
 - o Development of a safe and effective HIV vaccine is the object of active investigation.
 - o Extensive animal work is ongoing, and clinical trials of candidate vaccines are under way in humans.

ICD-9-CM

- 042 Human immunodeficiency virus [HIV] disease
- V08 Asymptomatic human immunodeficiency virus [HIV] infection status (includes HIV positive, not otherwise specified)

See Also

- Anal Cancer
- Approach to Weight Loss
- Candidiasis
- CNS Lymphoma
- Cryptococcosis
- Cryptosporidiosis
- Cytomegalovirus Infections
- Disorders of Smell
- Herpes Simplex Virus Infections
- Microsporidiosis
- Myocarditis
- Nontuberculous Mycobacterial Infections
- Non-Hodgkin Lymphomas
- Pneumocystis Infections

- Sclerosing Cholangitis
- Toxoplasmosis
- Tuberculosis Associated with HIV/AIDS

Internet Sites

- Professionals
 - AIDS Info
U.S. Department of Health and Human Services
- Patients
 - AIDS
MedlinePlus
 - AIDS and Infections
MedlinePlus

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